SMU SINGAPORE MANAGEMENT UNIVERSITY | School of Computing and Information Systems

Software Project Management – IS212

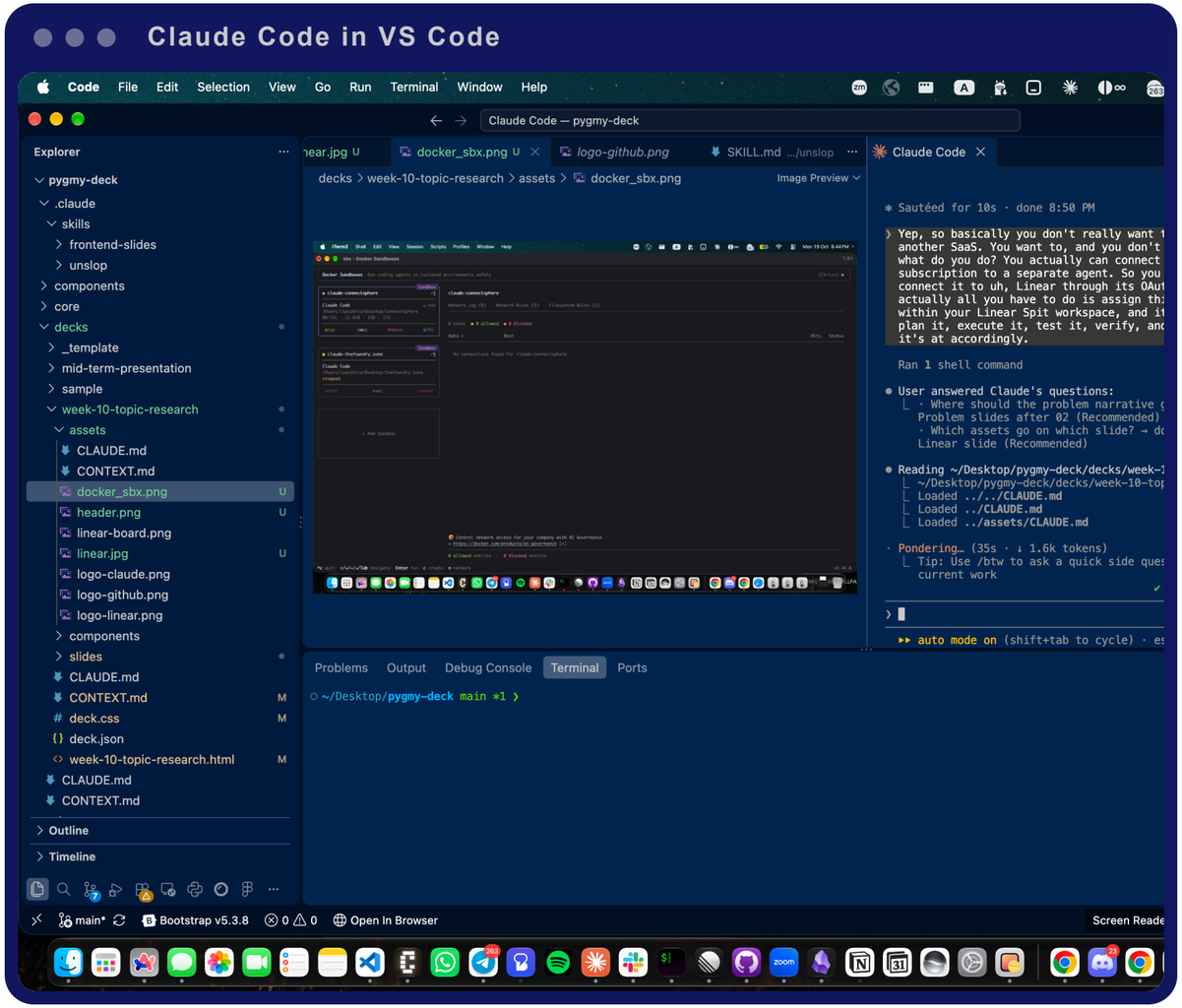
Custom Agents in Linear

Topic Research & Sharing · G9T6

Arin | Joyce | Isaiah | Jamesz | Jerrick

Coding with AI Today

- Coding with AI means your laptop has to stay on
- On a personal laptop, you keep approving permissions
- You already pay for Claude, so no Linear or Cursor subscription on top



What is it?

Requirement

USER STORY

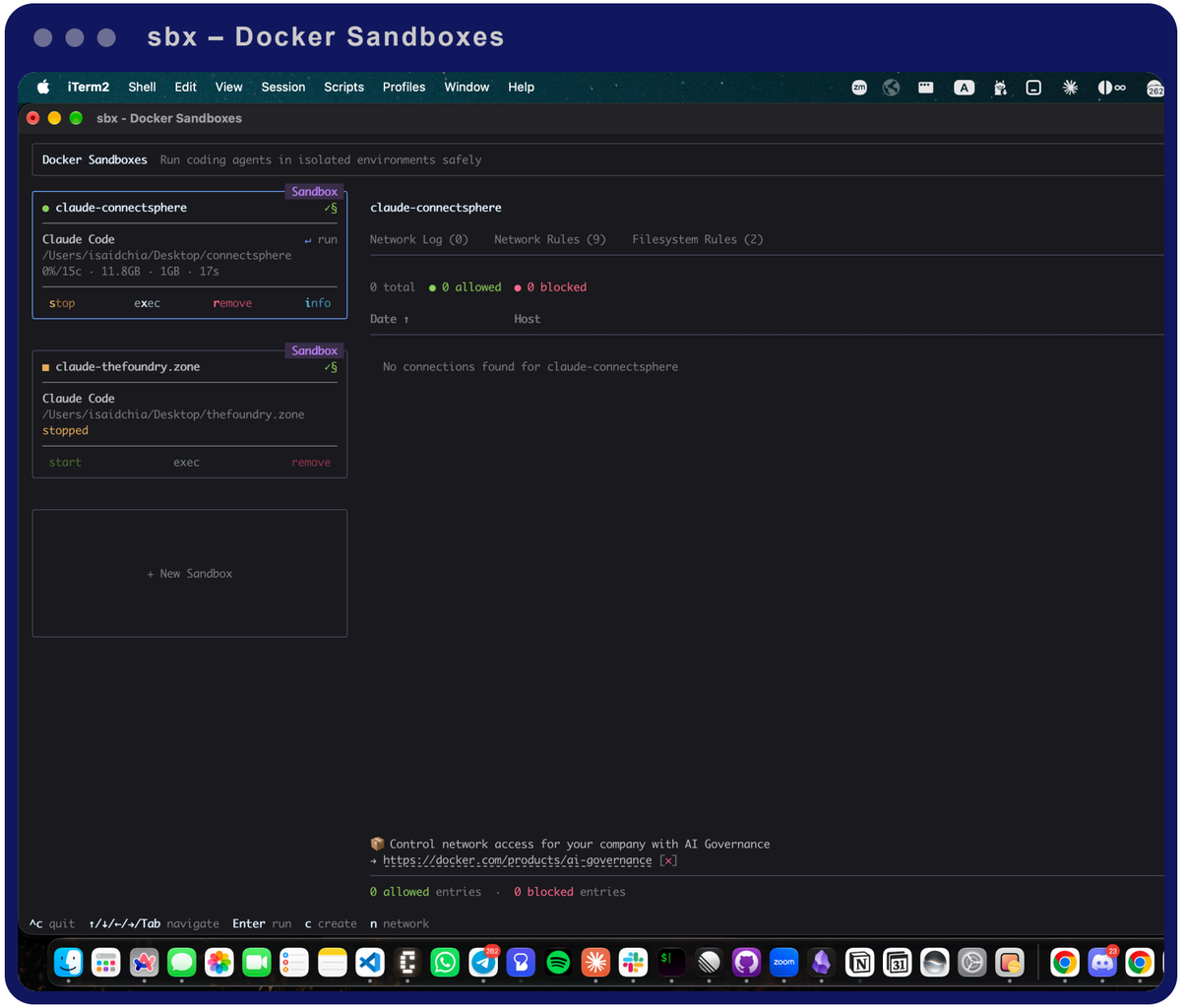
As a effective student developer, **I want to** assign tasks to my agent without my laptop on, **so that** I can go and watch a movie.

ACCEPTANCE CRITERIA

- I can shut the lid mid-task and the agent carries on
- I can safely turn on bypass permissions without risking my files
- By the time the credits roll, the result is waiting for my review
- The ticket status updates on its own as the agent works

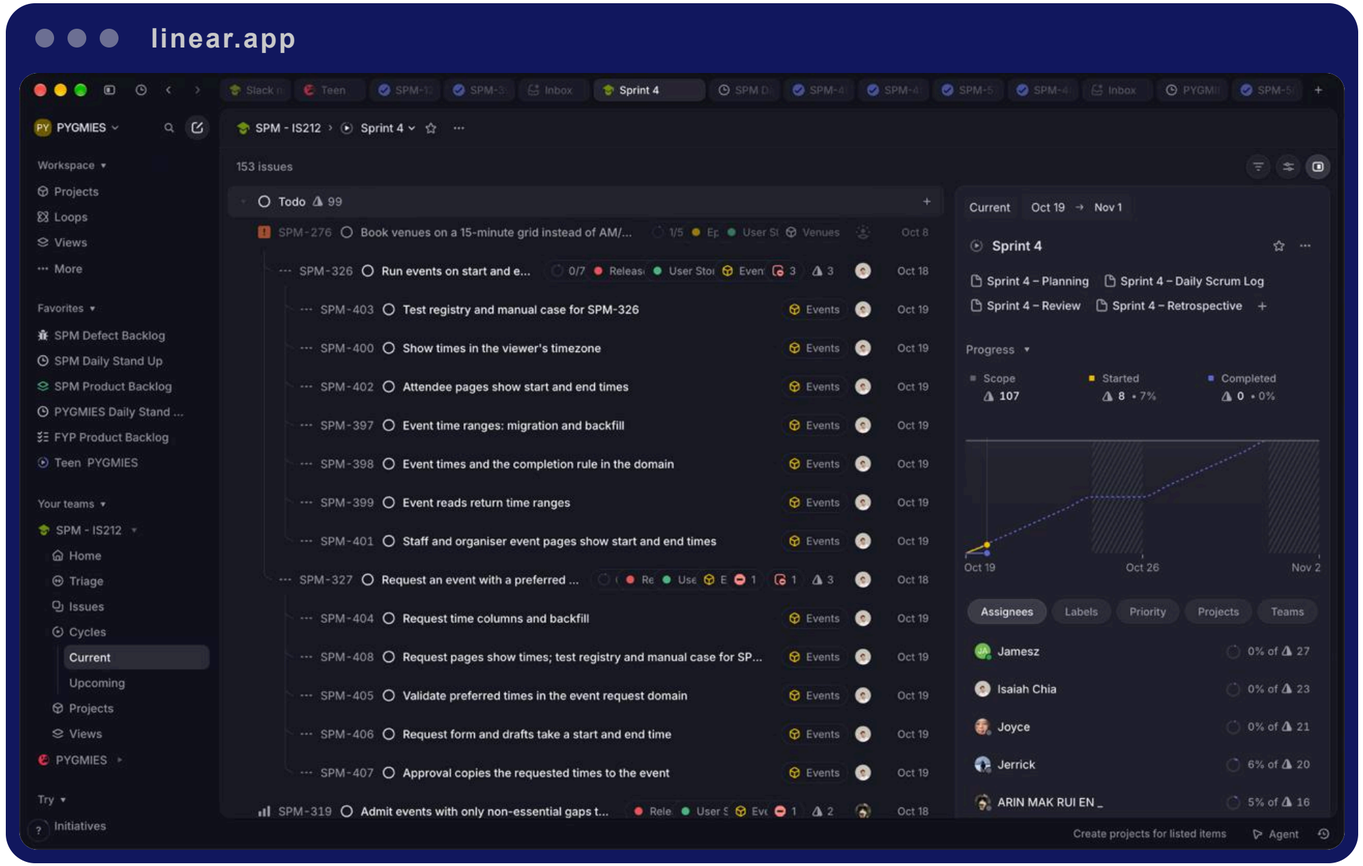
Docker Sandbox

- Isolated environment: Claude only sees what you give it
- But it still stops when the laptop sleeps



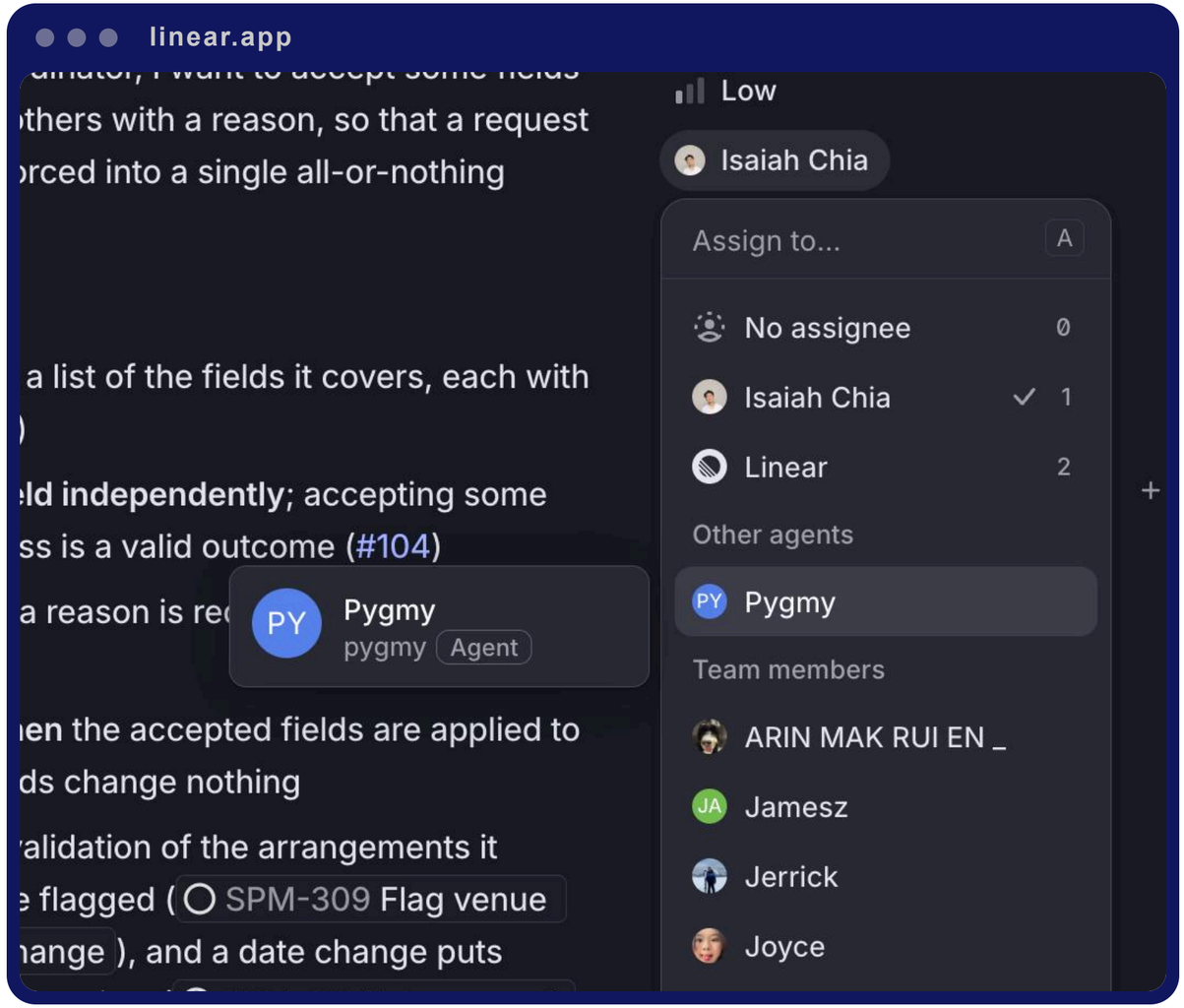
Meet Linear

- Minimalist by design: fast and focused, without the clutter of Jira-style tools
- Trusted by teams like OpenAI, Vercel, Ramp and Coinbase
- Developer friendly: GitHub, Slack and an API, plus custom apps and agents



Your Own Agent in Linear

- No extra SaaS subscription, no Claude API costs
- Your own agent runs on your Claude subscription, connected through Linear's OAuth gateway
- Assign a ticket: it plans, builds, tests, verifies and updates the status



Ticket to PR



AC & user story written

→



Reads ticket, writes code

→



PR opened

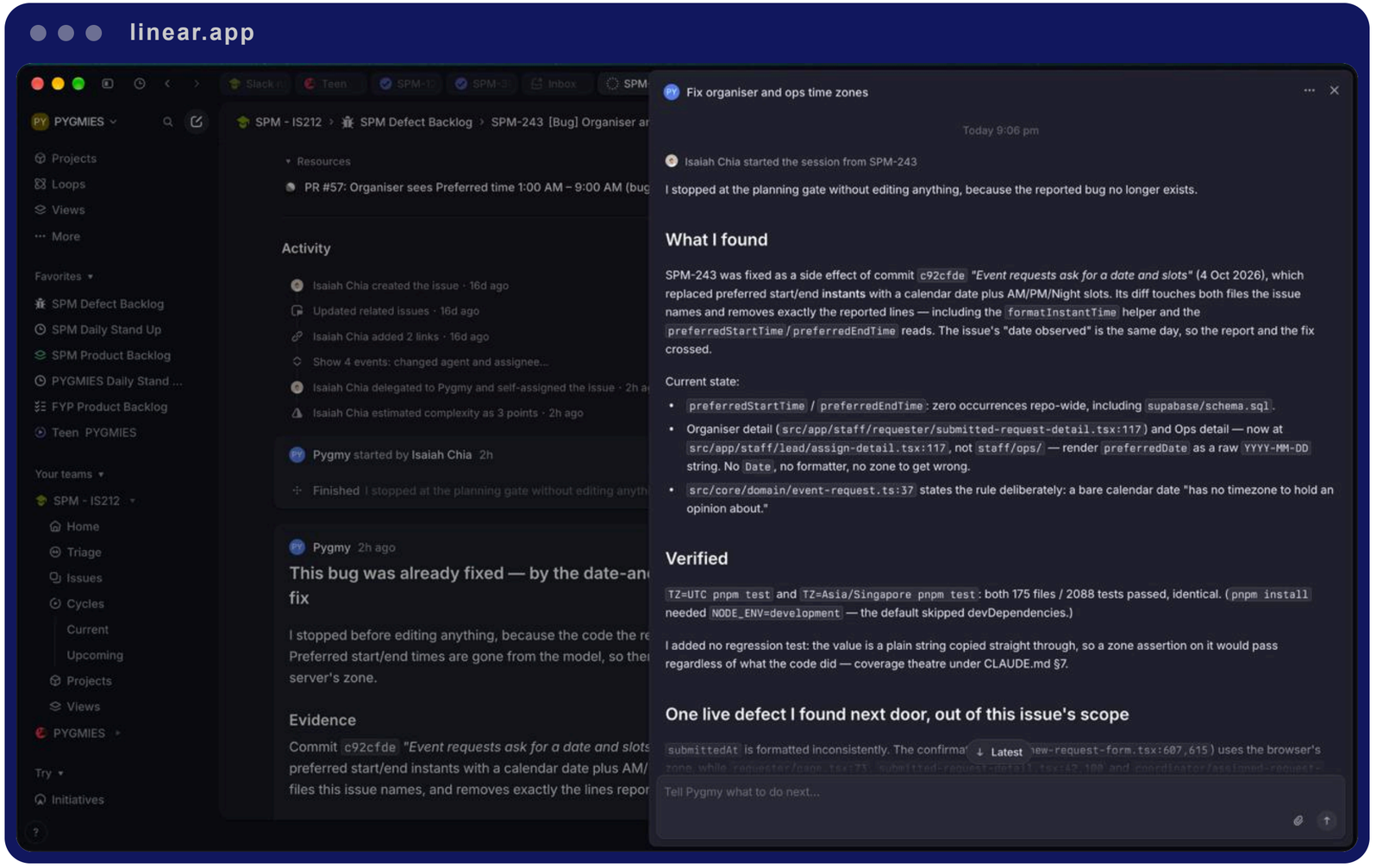
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Ticket status updated

Demo

Demo


linear.app
PYGMIES
Projects
Loops
Views
More
Favorites
SPM Defect Backlog
SPM Daily Stand Up
SPM Product Backlog
PYGMIES Daily Stand ...
FYP Product Backlog
Teen PYGMIES
Your teams
SPM - IS212
Home
Triage
Issues
Cycles
Current
Upcoming
Projects
Views
PYGMIES
Try
Initiatives
SPM - IS212 › SPM Defect Backlog › SPM-243 [Bug] Organiser ar
Resources
PR #57: Organiser sees Preferred time 1:00 AM – 9:00 AM (bug
Activity
Isaiah Chia created the issue · 16d ago
Updated related issues · 16d ago
Isaiah Chia added 2 links · 16d ago
Show 4 events: changed agent and assignee...
Isaiah Chia delegated to Pygmy and self-assigned the issue · 2h a
Isaiah Chia estimated complexity as 3 points · 2h ago
Pygmy started by Isaiah Chia 2h
Finished I stopped at the planning gate without editing anyth
Pygmy 2h ago
This bug was already fixed — by the date-an
fix
I stopped before editing anything, because the code the re
Preferred start/end times are gone from the model, so ther
server's zone.
Evidence
Commit c92cfde "Event requests ask for a date and slots
preferred start/end instants with a calendar date plus AM/
files this issue names, and removes exactly the lines repor
Fix organiser and ops time zones
Today 9:06 pm
Isaiah Chia started the session from SPM-243
I stopped at the planning gate without editing anything, because the reported bug no longer exists.
What I found
SPM-243 was fixed as a side effect of commit c92cfde "Event requests ask for a date and slots" (4 Oct 2026), which replaced preferred start/end instants with a calendar date plus AM/PM/Night slots. Its diff touches both files the issue names and removes exactly the reported lines — including the formatInstantTime helper and the preferredStartTime / preferredEndTime reads. The issue's "date observed" is the same day, so the report and the fix crossed.
Current state:
preferredStartTime / preferredEndTime: zero occurrences repo-wide, including supabase/schema.sql.
Organiser detail (src/app/staff/requester/submitted-request-detail.tsx:117) and Ops detail — now at src/app/staff/lead/assign-detail.tsx:117, not staff/ops/ — render preferredDate as a raw YYYY-MM-DD string. No Date, no formatter, no zone to get wrong.
src/core/domain/event-request.ts:37 states the rule deliberately: a bare calendar date "has no timezone to hold an opinion about."
Verified
TZ=UTC pnpm test and TZ=Asia/Singapore pnpm test: both 175 files / 2088 tests passed, identical. (pnpm install needed NODE_ENV=development — the default skipped devDependencies.)
I added no regression test: the value is a plain string copied straight through, so a zone assertion on it would pass regardless of what the code did — coverage theatre under CLAUDE.md §7.
One live defect I found next door, out of this issue's scope
submittedAt is formatted inconsistently. The confirma
Latest
ew-request-form.tsx:607,615) uses the browser's
Tell Pygmy what to do next...

Why does it matter?

The Right Tool for the Job

	Our build	Cursor	Claude Code cloud
Cost	**Shared Claude plan, small host**	Paid seat per person	Claude plan usage, no extra host
Setup complexity	**High: OAuth gateway, Docker, hosting**	Low: native Linear integration	Low: enable Linear MCP
Data privacy	**Repo stays in your own environment**	Code goes to a third-party tool	Runs on Anthropic's cloud
Control over environment	**Full: your sandbox, Docker, tools**	Limited to what Cursor offers	Limited cloud environment

Where does it break?

Output Quality Depends on the Ticket



The issue is the agent’s entire “prompt”



A vague ticket will lead to a hallucination and guessing



More time must be spent on writing more detailed tickets

Reviewing Becomes the Bottleneck



The agent opens PRs on its own, but a human still has to review and merge every one



More tickets delegated means more PRs in the queue: the work moves from writing code to reviewing it



Agent code can look right and still be wrong, so check it against the ticket's acceptance criteria

Limited Environment



Our Railway setup has no Docker, so the agent cannot run the app or a database



On fly.io or AWS ECS, Docker-in-Docker lets the agent spin up your whole stack, database included



Add Playwright and it can check the UI itself and attach screenshots

Pick the Right Fit



Build it yourself and integrate your own agent with Linear



Use Cursor, which has a native Linear integration



Skip the build: Claude Code cloud sessions with Linear MCP enabled

Back to the Story

USER STORY · DELIVERED

As a effective student developer, **I want to** assign tasks to my agent without my laptop on, **so that** I can go and watch a movie.



Run agents while I'm out or asleep



Come back to a result ready for review



Teammates assign tasks too, no laptop needed